

STAT 412 HW07 — Question 4

Problem

A research report claims that mice with an average life span of 32 months will live to be about 41 months old when 40% of the calories in their diet are replaced by vitamins and protein. Is there any reason to believe that $\mu < 41$ if 68 mice that are placed on this diet have an average life of 40 months with a standard deviation of 3.4 months? Use a P-value in your conclusion.

Solution

The question asks whether there is reason to believe $\mu < 41$, so the alternative is left-tailed:

$$H_0 : \mu = 41, \quad H_a : \mu < 41.$$

This is choice E.

Using the visible screenshot values,

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{40 - 41}{3.4/\sqrt{68}} \approx -2.4254.$$

Rounded to two decimals:

$$t = -2.43.$$

With $df = 67$, the exact left-tail P-value is about 0.009. From the t table, this falls in the range

$$0.0075 < P\text{-value} < 0.01.$$

Because the P-value is less than 0.05, reject H_0 .

Final Answer

Choice E: $H_0 : \mu = 41$; $H_a : \mu < 41$. $t = -2.43$; $0.0075 < P\text{-value} < 0.01$ (exact about 0.009).
Reject H_0 ; there is sufficient evidence that the mice will live to be less than 41 months old.